

SOLO® ADSS Short-Span, Loose Tube, Gel-Filled Cable 48 F, Single-mode (OS2)

CORNING

Part Number: 048EN4-T4S01A20

Corning SOLO® ADSS short-span cables are all-dielectric, self-supporting (ADSS) cables designed for easy and economical one-step installation in campus backbones with self-supporting installations where metallic messengers cannot be used. The loose tube design provides stable performance over a wide temperature range and is compatible with any telecommunications-grade optical fiber. The economical single-jacket design can span distances up to 600 ft in NESC light conditions, 500 ft in NESC medium conditions and 300 ft in NESC heavy conditions (see sag and tension chart for details). This cable incorporates innovative waterblocking materials, eliminating the need for traditional flooding compound and providing efficient and craft-friendly cable preparation. While the concentric, self-supporting cable design allows easy, one-step installation using standard hardware and installation methods, the SZ-stranded, loose tube design isolates optical fibers from installation and environmental rigors and facilitates mid-span access. The ADSS optical cables are also available with a proprietary track-resistant polyethylene (TRPE) jacket suitable for installation in electric field potentials up to 25 kV.

Features and Benefits

Loose tube design

Mechanical ruggedness and environmental durability

Self-supporting

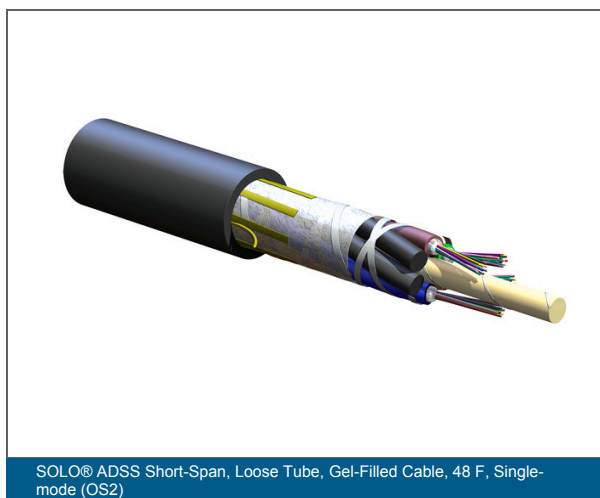
Easy, one-step installation

Track-resistant polyethylene (TRPE) jacket available

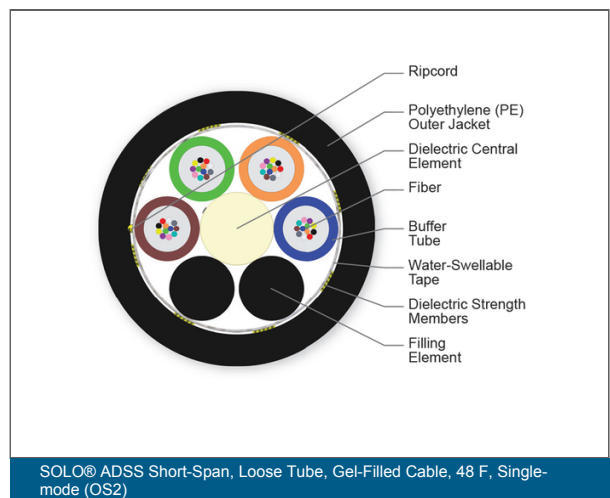
Suitable for installations up to 25 kV electric field potential

Innovative waterblocking design

Provides efficient and craft-friendly cable preparation



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Specifications

General Specifications

Cable Type	Loose Tube
Environment	Outdoor
Product Type	Self-Supporting, ADSS
Fiber Category	Single-mode (OS2)
Cable geometry	Other

Standards

RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Approvals and Listings	RDUP 7 CFR 1755.900 (formerly RUS)
Common Installations	Outdoor self-supporting aerial
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions

Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-40 °C to 70 °C (-40 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design

Central Element	Dielectric
Fiber Count	48
Number of Ripcords	1
Buffer Tube Color Coding, Layer 1	Blue, Orange, Green, Brown
Outer Jacket Color	Black
Outer Jacket Material	Polyethylene (PE)
Tensile Strength Elements and/or Armoring - Layer 1	Dielectric strength members
Buffer Tube Diameter	2.5 mm (0.1 in)

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Cable Design

Number of Active Tubes	4
Number of Filling Elements	2
Number of Tube Positions	6
Tape	Water-swellable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12

Mechanical Specifications

Nominal Outer Diameter	10.9 mm (0.43 in)
Min. Bend Diameter Installation	328 mm (12.91 in)
Min. Bend Diameter Operation	218 mm (8.58 in)

Optical Characteristics

Fiber Code	E
Performance Option Code	01
Fiber Category	OS2
Fiber Type	Single-mode (OS2) / 250 µm
Fiber Name	Single-mode (OS2)
Maximum Attenuation	0.4 dB/km / 0.4 dB/km / 0.3 dB/km
Wavelengths	1310 nm / 1383 nm / 1550 nm
Fiber Compliance	ITU-T G.652.D

Dimensions

Cable Weight	92 kg/km (61.82 lb/1000 ft)
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Installation Characteristics								
	Initial Installation		NESC Light		NESC Medium		NESC Heavy	
Span	SAG	Tension	SAG	Tension	SAG	Tension	SAG	Tension
Fiber Count 48								
15 m (49.2 ft)	1 %	171 N (38.4 lbf)	0.4 %	485 N (109 lbf)	1.7 %	690 N (155.1 lbf)	2.4 %	1,024 N (230.2 lbf)
30 m (98.4 ft)	1 %	343 N (77.1 lbf)	0.5 %	846 N (190.2 lbf)	2 %	1,173 N (263.7 lbf)	2.9 %	1,705 N (383.3 lbf)
46 m (150.9 ft)	1 %	514 N (115.5 lbf)	0.5 %	1,167 N (262.4 lbf)	2.2 %	1,594 N (358.4 lbf)	3.2 %	2,292 N (515.3 lbf)
61 m (200.1 ft)	1 %	685 N (154 lbf)	0.5 %	1,463 N (328.9 lbf)	2.3 %	1,980 N (445.1 lbf)	3.5 %	2,827 N (635.5 lbf)
76 m (249.3 ft)	1 %	857 N (192.7 lbf)	0.6 %	1,744 N (392.1 lbf)	2.5 %	2,342 N (526.5 lbf)	3.7 %	3,326 N (747.7 lbf)
91 m (298.5 ft)	1 %	1,028 N (231.1 lbf)	0.6 %	2,014 N (452.8 lbf)	2.6 %	2,687 N (604.1 lbf)	3.9 %	3,799 N (854 lbf)
107 m (351 ft)	1 %	1,199 N (269.6 lbf)	0.6 %	2,274 N (511.2 lbf)	2.7 %	3,019 N (678.7 lbf)	4.1 %	4,252 N (955.9 lbf)
122 m (400.2 ft)	1 %	1,371 N (308.2 lbf)	0.6 %	2,527 N (568.1 lbf)	2.8 %	3,339 N (750.6 lbf)	4.2 %	4,688 N (1,053.9 lbf)
137 m (449.4 ft)	1 %	1,542 N (346.7 lbf)	0.6 %	2,774 N (623.6 lbf)	2.9 %	3,650 N (820.5 lbf)	4.4 %	5,111 N (1,149 lbf)
152 m (498.6 ft)	1 %	1,713 N (385.1 lbf)	0.7 %	3,016 N (678 lbf)	3 %	3,954 N (888.9 lbf)	4.5 %	5,523 N (1,241.6 lbf)
168 m (551 ft)	1 %	1,885 N (423.8 lbf)	0.7 %	3,254 N (731.5 lbf)	3 %	4,251 N (955.7 lbf)		
183 m (600.2 ft)	1 %	2,056 N (462.2 lbf)	0.7 %	3,488 N (784.1 lbf)	3.1 %	4,543 N (1,021.3 lbf)		
198 m (649.4 ft)	1 %	2,227 N (500.6 lbf)	0.7 %	3,719 N (836.1 lbf)	3.2 %	4,829 N (1,085.6 lbf)		
213 m (698.6 ft)	1 %	2,399 N (539.3 lbf)	0.7 %	3,947 N (887.3 lbf)	3.2 %	5,111 N (1,149 lbf)		
229 m (751.1 ft)	1 %	2,570 N (577.8 lbf)	0.7 %	4,172 N (937.9 lbf)	3.3 %	5,388 N (1,211.3 lbf)		

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Installation Characteristics								
	Initial Installation		NESC Light		NESC Medium		NESC Heavy	
Span	SAG	Tension	SAG	Tension	SAG	Tension	SAG	Tension
244 m (800.3 ft)	1 %	2,742 N (616.4 lbf)	0.7 %	4,394 N (987.8 lbf)				
259 m (849.5 ft)	1 %	2,913 N (654.9 lbf)	0.7 %	4,615 N (1,037.5 lbf)				
274 m (898.7 ft)	1 %	3,084 N (693.3 lbf)	0.7 %	4,834 N (1,086.7 lbf)				
290 m (951.2 ft)	1 %	3,256 N (732 lbf)	0.8 %	5,051 N (1,135.5 lbf)				
305 m (1,000.4 ft)	1 %	3,427 N (770.4 lbf)	0.8 %	5,266 N (1,183.8 lbf)				
320 m (1,049.6 ft)	1 %	3,598 N (808.9 lbf)	0.8 %	5,479 N (1,231.7 lbf)				



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